AQA 3.2.4 Properties of Period 3 elements and their oxides

AQA A-level Chemistry · Revision questions · Sheet 1 of 6

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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18 marks · about 25 minutes

Key information

- Na_2O and MgO are giant ionic and basic: $Na_2O + H_2O \rightarrow 2NaOH$ (pH 12–14); $MgO + H_2O \rightarrow Mg(OH)_2$ (pH 9–10)
- Al_2O_3 is insoluble and amphoteric; SiO_2 is insoluble, giant covalent and acidic (reacts with hot conc NaOH)
- P_4O_{10}, SO_2, SO_3 are simple molecular and acidic: $P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$; $SO_2 \rightarrow H_2SO_3$; $SO_3 \rightarrow H_2SO_4$
- Melting points: giant ionic and giant covalent oxides are high; simple molecular oxides are low (weak van der Waals forces)
- $Mg + H_2O(g) \rightarrow MgO + H_2$ (steam); $2Na + 2H_2O \rightarrow 2NaOH + H_2$

Section A: Short-answer questions

01 Write an equation for the reaction of sulfur trioxide with potassium hydroxide solution.

[1 mark]

02 Give an equation for the reaction of sodium oxide with water. Suggest the pH of the solution formed.

[2 marks]

03 The oxides P_4O_{10} and SO_2 both react with water to form acid molecules. Which gives the correct shape around the central atom for each acid molecule that is formed? (molecule from P_4O_{10} / molecule from SO_2)

[1 mark]

A pyramidal / pyramidal
B pyramidal / tetrahedral
C tetrahedral / pyramidal
D tetrahedral / tetrahedral

04 Magnesium burns in oxygen. Give an equation for this reaction and state one observation.

[2 marks]

05 Give an equation for the reaction of magnesium oxide with hydrochloric acid.

[1 mark]

06 Identify the oxide that could react with water to form a solution with pH = 2

[1 mark]

A Al_2O_3
B Na_2O
C SiO_2
D SO_2

Section B: Longer questions

07 This question is about oxides and hydroxides of Period 3 elements.

07.1 An element in Period 3 forms an oxide that is insoluble in water. This oxide reacts with sulfuric acid and with aqueous potassium hydroxide. Give the formula for this oxide. Give an equation for the reaction of this oxide with sulfuric acid.

[2 marks]

07.2 Give the formula of a hydroxide of an element in Period 3 used in medicine.

[1 mark]

08 Explain, in terms of crystal structure and bonding, why silicon(IV) oxide has a higher melting point than phosphorus(V) oxide.

[4 marks]

09 A sample of the highest oxide of phosphorus was prepared in a laboratory. Describe a method for determining the melting point of the sample. State how the result obtained could be used to evaluate its purity.

[3 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	$SO_3 + 2KOH \rightarrow K_2SO_4 + H_2O$	*Source: 7405/1 Jun 2017 Q06.5* Allow $SO_3 + KOH \rightarrow KHSO_4$ Allow ionic: $SO_3 + 2OH^- \rightarrow SO_4^{2-} + H_2O$	1
02	$Na_2O + H_2O \rightarrow 2NaOH$ pH 12–14	*Source: Practice question (written for this set)*	2
03	**C**	*Source: 7405/3 Jun 2025 Q31*	1
04	$2Mg + O_2 \rightarrow 2MgO$ bright white light / flame **or** white powder / solid	*Source: Practice question (written for this set)* Do not allow ppt	2
05	$MgO + 2HCl \rightarrow MgCl_2 + H_2O$	*Source: Practice question (written for this set)*	1
06	**D**	*Source: 7405/1 Jun 2018 Q05.4* Original is tick-one-box	1
07.1	Al_2O_3 $Al_2O_3 + 3H_2SO_4 \rightarrow Al_2(SO_4)_3 + 3H_2O$	*Source: 7405/1 Nov 2021 Q03.4–03.5* Allow $Al_2O_3 + 6H^+ \rightarrow 2Al^{3+} + 3H_2O$	2
07.2	$Mg(OH)_2$		1
08	M1 SiO_2 is macromolecular / giant covalent M2 strong covalent bonds (between atoms) need a lot of energy to break M3 P_4O_{10} is molecular / simple molecular M4 weak van der Waals forces between molecules break easily	*Source: 7405/1 Nov 2021 Q03.3* Do not allow giant, giant atomic or giant ionic for M1	4
09	M1 sample in suitable melting point apparatus (e.g. capillary in oil bath / Thiele tube) M2 heat slowly / gradually M3 lower melting point / broad range indicates impurities **or** sharp melting point close to data book value indicates purity	*Source: 7405/3 Jun 2018 Q02.5* Do not allow water bath	3